Default Project 1 - DD2424*

Explore Transfer Learning

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Abstract

This study investigates transfer learning strategies for fine-grained image classification using the Oxford-IIIT Pet Dataset. Pre-trained ResNet18 and ResNet34 architectures were adapted to a 37-class breed classification task. Two transfer learning strategies were evaluated: simultaneous fine-tuning of multiple layers and gradual unfreezing. Both approaches exceeded the project target of 90% validation accuracy, with gradual unfreezing providing more stable optimization in low-data settings. To further study robustness under limited supervision, experiments were conducted using stratified 10% and 1% subsets of the training data. In the 1% data regime, MixUp augmentation acted as a strong regularizer and improved validation accuracy by +14.28%. Addressing class imbalance revealed severe data leakage caused by naive minority over-sampling before validation splitting. In addition, deeper architectures, catastrophic forgetting and representation learning were investigated through fine-tuning, selective layer freezing experiments and t-SNE feature visualizations. These experiments revealed how transfer learning alters internal feature representations and affects knowledge retention.

1 Introduction

Transfer learning has become a central technique in computer vision, enabling deep neural networks pre-trained on large datasets to be adapted to new tasks with limited labeled data. This is especially valuable in fine-grained classification, where visually similar classes and small datasets make training from scratch prone to overfitting. This project investigates transfer learning using the Oxford-IIIT Pet Dataset, where a pre-trained ResNet18 is first evaluated via linear probing before comparing two fine-tuning strategies: simultaneous layer fine-tuning and gradual unfreezing. Experiments are further extended to limited-data settings, class imbalance, deeper architectures, catastrophic forgetting, and MixUp augmentation. To better understand how transfer learning affects internal feature representations, t-SNE visualizations and k-nearest neighbor analysis are also performed.

2 Related Work

This study builds upon the findings of Kensert et al. (1), who highlight the efficacy of transfer learning in scenarios where annotated data is limited. Their work directly validates our experimental focus on analyzing fine-tuning performance across limited training subsets, while also providing a justification for our extended evaluation of deeper architectures.

For the foundational pipeline architecture and initial system verification, this work builds upon the established computer vision workflows demonstrated in the official fastai documentation (2). Following this framework allowed for rigorous sanity checking of data loading and model initialization, ensuring a stable and reproducible baseline before implementing our extended transfer learning strategies.

*Code repository available at: `https://github.com/WilliamBjornram/DD2424_Project.git`

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3 Data

The **Oxford-IIIT Pet Dataset** is used for evaluation. It contains 7,349 images across 37 pet breeds, with approximately 200 images per class. The dataset is challenging due to high intra-class variation and fine-grained inter-class differences.

To evaluate model robustness under severe data constraints, two stratified subsets were created:

- **1.0% Regime:** Approximately 2 images per breed, representing extreme data scarcity.
- **10.0% Regime:** Approximately 20 images per breed, representing moderate data scarcity.

Stratification ensures that all 37 classes remain uniformly represented across splits. For preprocessing, all images are resized to 256×256 pixels, center-cropped to $224 \times 224 \times 3$ and normalized using standard ImageNet channel statistics.

4 Methods

Before running the experiments, a sanity check was performed to ensure that the data pipeline and training setup worked correctly. For this initial check, the task was simplified to a binary classification problem: distinguishing between cats and dogs. Using a pre-trained ResNet18 model, the convolutional layers were frozen and only a newly added final classification layer was trained for three epochs. Following the successful sanity check, the pipeline was scaled to the full multi-class problem, incorporating all 37 breeds. This step served to establish a baseline performance and verify that the model architecture could handle the increased complexity of fine-grained classification before proceeding to further architectural experiments.

4.1 Fine-tuning Strategies

To optimize model performance, two distinct strategies were implemented to adapt the pre-trained weights to the dataset.

In **Strategy 1 (Independent Fine-tuning)**, each level of unfreezing was treated as a separate experiment. For each depth l ($l \in \{1, 2, 3\}$), a fresh model instance was initialized from the pre-trained state. The final l layer groups were unfrozen and trained independently. To account for the increasing number of trainable parameters, the training duration was scaled accordingly, utilizing 2, 4 and 6 epochs for the respective unfreezing depths.

In **Strategy 2 (Gradual Unfreezing)**, a single model instance was trained cumulatively. Layer groups were progressively unfrozen in stages ($l = 1$ to 3). By retaining optimized weights from previous stages, the model refined features iteratively. A fixed schedule of two epochs per stage ensured stable adaptation and standardized the comparison against Strategy 1 while mitigating overfitting.

4.2 Limited Data Experiments

To evaluate performance under constrained supervision, experiments were run on 100%, 10% and 1% subsets of the dataset. To mitigate the severe overfitting risk in these low-data regimes, data augmentation and varying L2 regularization (wd) were also tested. Results show that extreme data scarcity significantly reduces generalization. However, gradual unfreezing consistently improves stability in low-data regimes by preserving pretrained features more effectively.

4.3 Class Imbalance

To simulate realistic data skews, the training data of all cat breeds was reduced to 20% of their original volume, while dog breeds were kept at full size. To evaluate the models performance `scikit-learn`'s macro-averaged and per-class F_1-scores were used rather than global accuracy, as standard accuracy metrics mask poor minority performance.

Two distinct approaches were investigated:

- Baseline training on imbalanced data using standard cross-entropy loss.

- Oversampling of minority classes before dataset splitting.

A critical architectural flaw was found during the over-sampling pipeline. Because duplicate samples are introduced before the data wrapper splits the files into distinct training and validation sets, identical image vectors are allocated to both splits simultaneously. This creates severe data leakage, where the validation set no longer represents unseen data. Consequently, giving a too optimistic result. To fix this the split would have to happen before the sampling begins, so the validation set is separated from the training data.

4.4 Deeper Network

To evaluate the impact of architectural depth, a pre-trained ResNet34 was compared against a ResNet18 baseline using initial linear probing followed by full fine-tuning. The fine-tuning phase compared Strategy 1 against Strategy 2. Models were trained using AdamW, the default optimizer in fastai, and L2 regularization was analyzed by testing various weight decay values to investigate the optimal settings.

4.5 Representation Analysis

To better understand how transfer learning alters the model's internal understanding of the data, a representation analysis was conducted. First, feature embeddings were extracted from the network's penultimate layer. An embedding (or feature map) is a high-dimensional mathematical vector that summarizes the complex visual characteristics of an image, such as textures, edges and anatomical shapes. To make these high-dimensional features human-readable, they were projected into a 2D space using t-SNE for a subset of 10 breeds. Additionally, a k-nearest neighbors ($k = 5$) analysis was established using cosine distance to measure the mathematical similarity between image embeddings in the validation set.

4.6 Catastrophic Forgetting

To evaluate catastrophic forgetting without class bias, a stratified test set of 20 images per breed (10% of the data) was isolated from the dataset (4,900 dogs/25 breeds; 2,400 cats/12 breeds), with the remainder split 80/20 for training and validation. A baseline ResNet18 trained on dogs for four epochs established foundational metrics. To induce forgetting, its convolutional layers were transferred to a cat-configured ResNet18 and fine-tuned on cats for four epochs using three strategies: full fine-tuning, unfreezing only the final layer group, and unfreezing the final two layer groups. Catastrophic forgetting was then quantified by evaluating these cat-trained models on the unseen dog test set, using overall accuracy and average F1 score to ensure a robust, class-balanced assessment.

4.7 Data Augmentation

To investigate the impact of sophisticated data regularization under varying degrees of sample scarcity, the MixUp augmentation was used during training. MixUp works by generating composite training images from linear combinations of random image pairs. Rather than training strictly on distinct, individual class images, the network is trained on "in-between" images where both the pixel data and the corresponding category labels are mixed proportionally. This regularizes the network by preventing it from becoming overconfident on noisy or scarce data.

To systematically evaluate generalization capacity models was tested against two constrained, stratified profiles representing 10.0% and 1.0% of the complete training dataset. For each data regime, three distinct architectural configurations were compared:

- **Baseline:** No augmentation.
- **Basic Augmentation:** Standard spatial augmentation (cropping, flipping).
- **Advanced Augmentation (MixUp):** MixUp + spatial augmentation.

5 Experiments

5.1 Baseline Performance

In the binary sanity check, training and validation losses converged to 0.0369 and 0.0226 respectively, with validation accuracy peaking at 99.3%. Following this initial validation, the linear probing setup was extended to the full multi-class problem using all 37 breeds. As expected, the higher complexity of distinguishing between 37 specific breeds resulted in a lower baseline validation accuracy. This drop in performance is expected when scaling to fine-grained classification without fine-tuning the underlying convolutional layers. Nevertheless, the network optimized successfully, as both the training loss and validation loss decreased consistently, dropping to 0.388 and 0.229, respectively, while the validation accuracy improved from an initial 86.7% to 92.6%. These results validate the high quality of the pre-trained feature representations, establishing a robust baseline for subsequent fine-tuning.

5.2 Fine-tuning Results

Analyzing the two different strategy resulted in a peak validation accuracy of 93.7% for strategy 1, with the training and validation losses minimizing to 0.259 and 0.222, respectively. Strategy 2 stabilized at a validation accuracy of 92.1%, with training and validation losses dropping to 0.296 and 0.229. Ultimately, both approaches successfully exceeded the $> 90\%$ target accuracy guideline.

5.3 Limited Data Results

Table 1: Validation accuracy across different data fractions, augmentation settings and L2 regularization (weight decay) for both transfer learning strategies.

			Validation Accuracy (%)	
Data Fraction	Augmentation	Weight Decay (wd)	Strategy 1	Strategy 2
100%	False	0	91.3	91.9
100%	True	0.01	90.5	90.9
100%	True	0.1	91.3	91.5
10%	False	0	55.1	82.3
10%	True	0.01	63.3	85.7
10%	True	0.1	64.6	81.0
1%	False	0	14.3	21.4
1%	True	0.01	0.0	28.6
1%	True	0.1	0.0	14.3

5.4 Class Imbalance Results

Table 2: Validation performance comparison highlighting the effects class imbalance and data leakage (see full table in Appendix).

Metric Group	**Uncompensated Baseline**	**Over-sampling**
Global Accuracy	85.00%	96.00%
Minority Cat Breeds		
Avg. Validation Support	$\sim$ 8 – 11	$\sim$ 30 – 50
Macro F_1-Score	0.74	**1.00**
Majority Dog Breeds		
Avg. Validation Support	$\sim$ 30 – 50	$\sim$ 30 – 50
Macro F_1-Score	0.88	0.89

The uncompensated baseline highlights distinct generalization drops among the underrepresented cat categories. While the undisturbed dog classes maintain high tracking metrics. This occurs because standard cross-entropy loss shifts decision boundaries to prioritize majority class optimization at the expense of minority groups.

When introducing the over-sampling strategy, the validation shows a global accuracy of 96% and a perfect 1.00 F_1-score across almost all cat classes. This is definitive proof of severe data leakage.

5.5 Deeper Network Results

To evaluate the impact of architectural depth, a pre-trained ResNet34 was tested (Table 3). Initial linear probing achieved 93.2% validation accuracy, outperforming the ResNet18 baseline due to its higher representational capacity for complex features. During fine-tuning, Strategy 2 reached a peak accuracy of 94.0%, outperforming Strategy 1. This demonstrates that progressively unfreezing layers safely preserves pre-trained weights, whereas simultaneously updating multiple layers from the start of training (Strategy 1) disrupts foundational representations. Finally, to manage overfitting risks in this larger architecture, L2 regularization (weight decay) was analyzed. Increasing weight decay to 0.1 or 0.05 degraded performance, indicating that the default AdamW regularization combined with baseline data augmentation was already sufficient.

Table 3: Performance comparison of the ResNet34 architecture.

Experimental Setup		
Model Architecture	Training Configuration	Validation Accuracy (%)
ResNet18	Linear Probing	92.6
ResNet34	Linear Probing	93.2
ResNet34	Strategy 1 (wd = 0.01)	92.2
ResNet34	Strategy 2 (wd = 0.01)	94.0
ResNet34	Strategy 2 (wd = 0.05)	92.9
ResNet34	Strategy 2 (wd = 0.10)	93.4

5.6 Representation Analysis

Figure 1a show that the pretrained embeddings initially form overlapping clusters, indicating weak class separation. However, after fine-tuning, the network successfully adapted its internal representations. Figure 1b display well-separated clusters for each breed, explaining the sharp increase in classification accuracy. This successful adaptation was further confirmed by the k-nearest neighbors analysis (see appendix A).

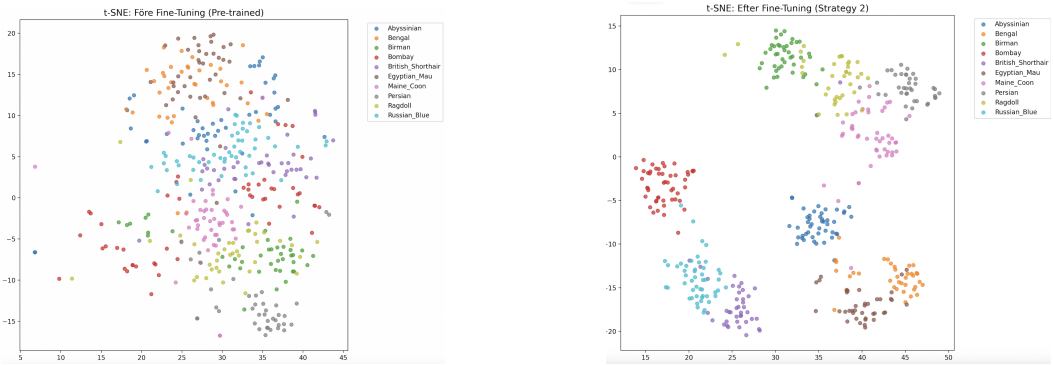


Figure 1: t-SNE visualization of feature embeddings for a subset of 10 breeds before (left) and after (right) fine-tuning.

5.7 Catastrophic Forgetting

A baseline model trained exclusively on dogs achieved a robust 93.80% test accuracy (F1: 0.9372) on this unseen set. However, when this model was subsequently fully fine-tuned on the cat dataset, its performance on the dog test set dropped significantly to 76.20% (F1: 0.7491). This severe degradation clearly demonstrates catastrophic forgetting, as the network overwrote previously learned dog features to optimize for cats. To mitigate this, partial fine-tuning was explored by freezing early layers at different depths during the cat training phase. Unfreezing only the final layer yielded no retention improvement, resulting in a test accuracy of 75.60% (F1: 0.7433). However, unfreezing the last two layers forced the model to retain its foundational feature extractors while still adapting successfully, which improved the retained dog test accuracy to 79.00% (F1: 0.7842).

Table 4: Evaluation of catastrophic forgetting on the isolated dog test set.

Training Phase	Frozen Layers	Test Accuracy (%)	F1 Score
Baseline (Trained on Dogs)	None	93.80	0.9372
Fine-tuned on Cats	None	76.20	0.7491
Fine-tuned on Cats	Last layer	75.60	0.7433
Fine-tuned on Cats	Last two layers	79.00	0.7842

5.8 Data Augmentation Results

The quantitative performance results across the two data scarcity levels are presented in Table 5. Under extreme data restrictions (1.0% regime), both the baseline and basic augmentation models reach a performance ceiling, obtaining an identical validation accuracy of 21.43%. With so few training samples, the ResNet18 model simply memorizes low-level pixel details rather than generalizable features. This overfitting is clearly reflected in very low training losses paired with low validation accuracy.

In contrast, the MixUp pipeline acts as a strong regularizer, boosting validation accuracy to 35.71%, an absolute improvement of +14.28%. The higher training loss observed during MixUp training shows that blending image pairs successfully prevents the network from memorizing raw data points, forcing it to learn smoother decision boundaries.

Table 5: Validation Accuracy (%) comparisons under varying regularization frameworks and stratified training data.

Data	**No Augmentation**	**Basic Augmentation**	**Adv. Augmentation (MixUp)**
1.0% Regime	21.43%	21.43%	**35.71%**
10.0% Regime	81.63%	**82.99%**	79.59%

When the training data is increased to the 10.0% regime, the performance trends shift. The Basic Augmentation model leads at 82.99%, slightly outperforming the MixUp model (79.59%). This outcome highlights a trade-off in optimization speed. Because MixUp trains on blended, soft target distributions rather than simple discrete classes, it creates a more complex optimization landscape. Within a short training of 30 epochs, basic spatial augmentations allow the model to converge quickly onto the clean images. Meanwhile, the MixUp pipeline requires a longer training time to fully optimize, which could explain its performance.

6 Conclusion

This study demonstrates transfer learning optimization for fine-grained classification under severe constraints. Gradual unfreezing enabled ResNet34 to preserve pre-trained weights and achieve 94.0% accuracy, while t-SNE confirmed distinct breed clustering. Under extreme data scarcity (1%), MixUp regularized training, boosting performance by 14.28% and freezing foundational layers mitigated catastrophic forgetting. Future work should focus on a detailed error analysis of the remaining 6% misclassifications to understand the model's absolute limits.

References

[1] Kensert, A., Harrison, P.J. & Spjuth, O. (2019) Transfer Learning with Deep Convolutional Neural Networks for Classifying Cellular Morphological Changes. *SLAS Discovery* 24, pp. 466–475.

[2] Howard, J. & Gugger, S. (2020) Fastai Computer Vision Tutorial. Available at: `https://docs.fast.ai/tutorial.vision.html` (Accessed: May 2026).

A Appendix: Class Imbalance Results

This appendix presents the complete, per-class classification reports extracted via `scikit-learn` for both the artificially skewed 20% cat-breed downsampled configuration (Table 6) and the over-sampled configuration showing data leakage (Table 7). Cat breeds are indicated by capitalized initial letters, while dog breeds are lowercased.

Table 6: Granular validation metrics for the Uncompensated Imbalanced Baseline experiment.

Breed Class Name	**Precision**	**Recall**	F_1**-Score**	**Support**
Abyssinian	0.89	0.80	0.84	10
Bengal	0.73	0.89	0.80	9
Birman	0.75	0.55	0.63	11
Bombay	1.00	0.71	0.83	7
British_Shorthair	0.88	0.70	0.78	10
Egyptian_Mau	0.67	0.67	0.67	6
Maine_Coon	1.00	0.60	0.75	10
Persian	0.86	0.86	0.86	7
Ragdoll	0.46	0.86	0.60	7
Russian_Blue	0.57	0.67	0.62	6
Siamese	0.86	0.75	0.80	8
Sphynx	1.00	0.78	0.88	9
american_bulldog	0.76	0.79	0.77	28
american_pit_bull_terrier	0.67	0.71	0.69	34
basset_hound	0.93	0.84	0.88	50
beagle	0.82	0.90	0.86	40
boxer	0.92	0.94	0.93	35
chihuahua	0.85	0.95	0.90	43
english_cocker_spaniel	0.87	0.93	0.90	42
english_setter	0.95	0.86	0.90	44
german_shorthaired	0.88	0.97	0.92	36
great_pyrenees	0.93	0.95	0.94	39
havanese	0.97	0.91	0.94	33

Table 7: Granular validation metrics for the Naive Over-sampled experiment highlighting the validation support inflation and artificial 1.00 scores due to data leakage prior to splitting.

Breed Class Name	**Precision**	**Recall**	**F_1-Score**	**Support**
Abyssinian	1.00	1.00	1.00	79
Bengal	1.00	1.00	1.00	61
Birman	1.00	1.00	1.00	79
Bombay	1.00	1.00	1.00	73
British_Shorthair	1.00	1.00	1.00	78
Egyptian_Mau	1.00	1.00	1.00	76
Maine_Coon	0.99	1.00	0.99	85
Persian	1.00	1.00	1.00	84
Ragdoll	1.00	1.00	1.00	86
Russian_Blue	1.00	1.00	1.00	75
Siamese	1.00	1.00	1.00	83
Sphynx	1.00	1.00	1.00	74
american_bulldog	0.80	0.82	0.81	44
american_pit_bull_terrier	0.84	0.84	0.84	45
basset_hound	0.85	0.92	0.88	36
beagle	0.93	0.84	0.88	31
boxer	0.83	0.95	0.89	41
chihuahua	0.91	0.83	0.87	36
english_cocker_spaniel	0.94	0.89	0.91	36
english_setter	0.91	0.95	0.93	42
german_shorthaired	0.92	0.92	0.92	39
great_pyrenees	0.97	1.00	0.99	34
havanese	0.90	0.90	0.90	40
accuracy			**0.96**	**1958**
macro avg	0.95	0.95	**0.95**	1958
weighted avg	0.96	0.96	**0.96**	1958

B Appendix: Representation Space, k-nearest

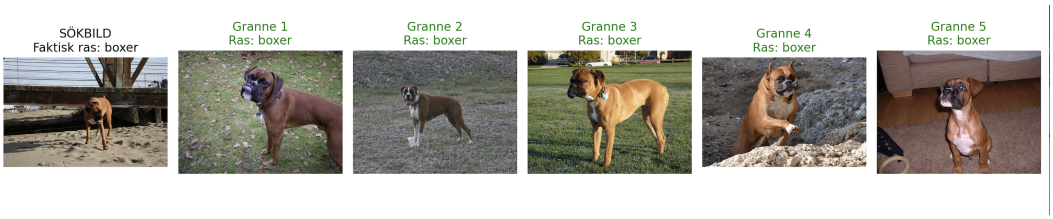
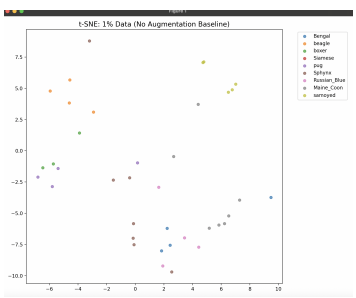


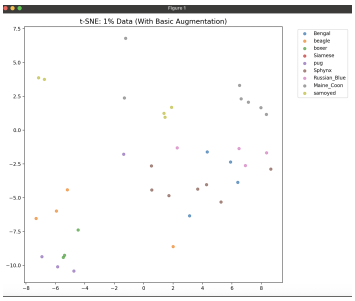
Figure 2: Qualitative k-NN image retrieval ($k = 5$) results. The network successfully matches five distinct images of the identical breed (*boxer*) across heavily varying lighting, background environments (grass, gravel, carpet) and dog poses, demonstrating clean semantic clustering in the representation space.

C Appendix: t-SNE Visualizations during extreme sacristy(1% Data)

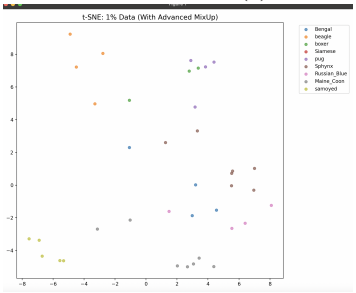
Figure 3 shows the t-sne plots corresponding to the reults from Table 5 in Section 5.8



(a) No Augmentation Baseline



(b) Basic Augmentation



(c) Advanced MixUp

Figure 3: Side-by-side 2D t-SNE manifold projections extracted from the final pooling layers of the three models under the 1% data footprint regime.